Weierstrass convergence Theorem.

Def. A sequence $f_n : G \to \mathbb{C}$ converges locally uniformly to f if: for every compact $K \subset G$,

$$\sup_{z \in K} |f_n(z) - f(z)| \to 0 \quad \text{as } n \to \infty.$$

Thm. Suppose that $f_n : G \to \mathbb{C}$ is holomorphic on G and $f_n \to f$ locally uniformly, then f is holomorphic. Moreover, $f_n' \to f'$ locally uniformly on G, (G open)

Proof. Given $z_0 \in G$, take $\delta > 0$ such that $\overline{B(z_0, r)} \subset G$. By the hypothesis, $f_n \to f$ uniformly on $\overline{B_\delta(z_0)}$, and so f is continuous on $B_\delta(z_0)$.

Let T be a triangular path inside $B_\delta(z_0)$. Since $f_n \to f$ uniformly,

$\int_T f_n(z)\, dz \to \int_T f(z)\, dz$. But, by the Cauchy-Goursat's theorem, each $n \in \mathbb{N}$, $\int_T f_n(z)\, dz = 0$, so $\int_T f(z)\, dz = 0$.

Now, by the Morera's theorem, f is Holomorphic on $B_\delta(z_0)$.

Since z_0 was arbitrary, so f is Holomorphic on G. To show the second assertion, let $K \subset G$ be a compact subset. Since $\mathbb{C}$ is T_4 space and G is open, there is an open U such that $K \subseteq U \subseteq \overline{U} \subseteq G$.

Now, for each $\delta > 0$, define

$$U_\delta := \{z \in U \mid \overline{D_\delta}(z) \subset U\} \text{ where } D_\delta = \{z' \in \mathbb{C} \mid |z' - z| < \delta\}.$$

Given $z \in U_\delta$, by the Cauchy's integral formula,

$$f'(z) - f_n'(z) = \frac{1}{2\pi i} \cdot \int_{C_\delta} \frac{f(w) - f_n(w)}{(w - z)^2}\, dw$$

where C_δ is a curve centered at z with radius $\delta > 0$. Therefore,

$$|f'(z) - f_n'(z)| \le \frac{1}{2\pi} \cdot \int_{C_\delta} \frac{|f(w) - f_n(w)|}{|w - z|^2} |dw| \le \frac{1}{2\pi} \cdot \frac{2\pi\delta}{\delta^2} \cdot \sup_{w \in C_\delta} |f(w) - f_n(w)|$$

$$\le \frac{1}{\delta} \cdot \sup_{w \in \overline{U}} |f(w) - f_n(w)| \to 0 \quad \text{as } n \to \infty.$$

Since the bound that tends to zero is independent of choice of z, Therefore, we can choose $\delta > 0$ small enough to $K \subset U_\delta$, so $f_n' \to f'$ uniformly. ☐